

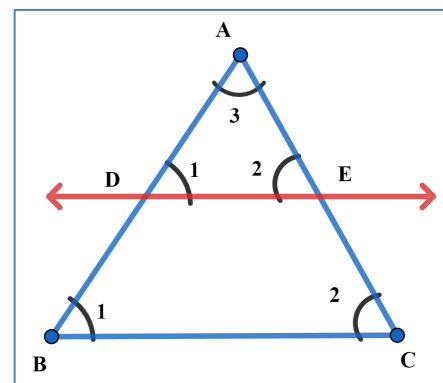
10 TH MATHEMATICS THEOREMS STUDY MATERIAL

THEOREM 1

THALES THEOREM OR BPT (BASIC PROPORTIONALITY THEOREM)

1. STATEMENT

- ❖ A **Straight line** drawn **Parallel** to a side of triangle intersecting the other two sides, **divides** the sides in the **Same ratio**.

2. GIVEN: In $\triangle ABC$ D and E is a Point on AB and AC3. TO PROVE: $\frac{AD}{DB} = \frac{AE}{EC}$ 4. CONSTRUCTION: Draw a line $DE \parallel BC$.

S. NO	STATEMENT	REASON
1.	$\angle ABC = \angle ADE = \angle 1$	Corresponding Angles Equal ($DE \parallel BC$)
2.	$\angle ACB = \angle AED = \angle 2$	
3.	$\angle DAE = \angle BAC = \angle 3$	Both Triangle a Common Angle.
4.	$\triangle ABC \sim \triangle ADE$ $\frac{AB}{AD} = \frac{AC}{AE}$ $\frac{AD + DB}{AD} = \frac{AE + EC}{AE}$ $1 + \frac{DB}{AD} = 1 + \frac{EC}{AE}$ $\frac{DB}{AD} = \frac{EC}{AE}$ $\frac{AD}{DB} = \frac{AE}{EC}$	By AAA Similarity Corresponding Sides Proportional Split AB and AC Using Points D and E Cancelling 1 on both sides Taking Reciprocals Hence Proved

THEOREM 2

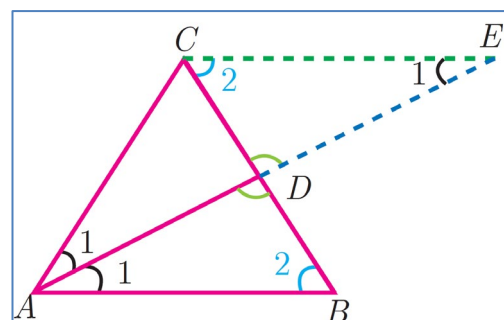
ABT (ANGLE BISECTOR THEOREM)

1. STATEMENT

- ❖ The **Internal Bisector** of an angle of a triangle divides the **Opposite side Internally** in the ratio of the **Corresponding sides Containing the Angle**.

2. GIVEN: In $\triangle ABC$ AD is the Internal Bisector3. TO PROVE: $\frac{AB}{AC} = \frac{BD}{CD}$

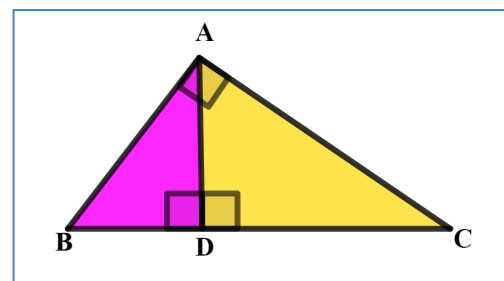
4. CONSTRUCTION: Draw a line Parallel to AB. Extend AD line Through C at E.



S. NO	STATEMENT	REASON
1.	$\angle AEC = \angle BAE = \angle 1$	Two parallel lines Cut a transversal. (Alternate Angles Equal)
2.	$\angle ABD = \angle ECD = \angle 2$	
3.	ΔACE is isosceles $AC = CE$ -----(1)	In ΔACE , $\angle CAE = \angle CEA$
4.	$\Delta ABD \sim \Delta ECD$ $\frac{AB}{CE} = \frac{BD}{CD}$ $\frac{AB}{AC} = \frac{BD}{CD}$	By AAA Similarity From (1) $AC = CE$ Hence Proved

THEOREM 3**PYTHAGORAS THEOREM (BAUDHAYANA THEOREM)****1. STATEMENT**

❖ In a Right angle Triangle, the Square on the Hypotenuse is equal to the sum of the Squares on the other Two sides.



2. GIVEN: In ΔABC $\angle A = 90^\circ$ is the Internal Bisector

3. TO PROVE: $AB^2 + AC^2 = BC^2$

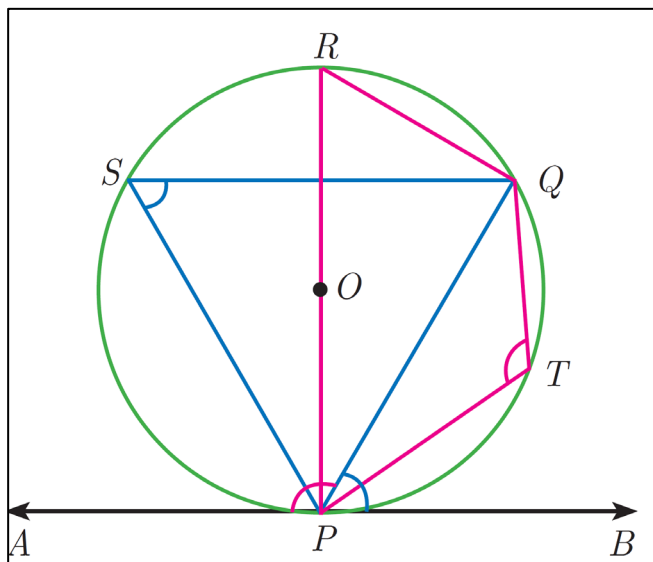
4. CONSTRUCTION: Draw $AD \perp BC$

S. NO	STATEMENT	REASON
1.	ΔABC and ΔDBA ($\angle B$ is Common) $\angle BAC = \angle BDA = 90^\circ$	Given $\angle BAC = 90^\circ$ and $\angle BDA = 90^\circ$
2.	$\therefore \Delta ABC \sim \Delta DBA$ $\frac{AB}{BD} = \frac{BC}{AB}$ $AB^2 = BC \times BD$ -----(1)	By AA Similarity
3.	ΔABC and ΔDAC ($\angle C$ is Common) $\angle BAC = \angle ADC = 90^\circ$	Given $\angle BAC = 90^\circ$ and $\angle BDA = 90^\circ$
4.	$\therefore \Delta ABC \sim \Delta DAC$ $\frac{BC}{AC} = \frac{AC}{DC}$ $AC^2 = BC \times DC$ -----(2)	By AA Similarity
Add (1) + (2) $\Rightarrow AB^2 + AC^2 = BC \times BD + BC \times DC$ $= BC[BD + DC]$ $= BC \times BC$ $AB^2 + AC^2 = BC^2$ $\therefore$ Hence Proved Theorem		

PREPARED AND TYPED BY
Y. SEENIVASAN. M.Sc, B.Ed (MATHS TEACHER)

THEOREM 4**ALTERNATE SEGMENT THEOREM (CHORD TANGENT THEOREM)****1. STATEMENT**

- ❖ If a line touches a circle and from the point of contact a chord is drawn, the angles between the tangent and the chord are respectively equal to the angles in the corresponding alternate segments.

**2. GIVEN:**

- ❖ A circle with Centre at O , tangent AB touches the circle at P and PQ is a chord. S and T are two points on the circle in the opposite sides of chord PQ .

3. TO PROVE (i) $\angle QPB = \angle PSQ$ and (ii) $\angle QPA = \angle PTQ$ **4. CONSTRUCTION:** Draw the diameter POR . Draw QR , QS and PS .

S. NO	STATEMENT	REASON
1.	$\angle RPB = 90^\circ$, $\angle RPQ + \angle QPB = 90^\circ$ --- (1)	Diameter RP is perpendicular to tangent AB .
2.	In $\triangle RPQ$, $\angle PQR = 90^\circ$ --- (2)	Angle in a semicircle is 90° .
3.	$\angle QRP + \angle RPQ = 90^\circ$ --- (3)	In a Right angled triangle, sum of the two acute angles is 90° .
4.	$\angle RPQ + \angle QPB = \angle QRP + \angle RFQ$ $\angle QPB = \angle QRP$ --- (4)	From (1) and (3)
5.	$\angle QRP = \angle PSQ$ --- (5)	Angles in the same segment are equal.
6.	$\angle QPB = \angle PSQ$ --- (6)	From (4) and (5) Hence (i) is proved.
7.	$\angle QPB + \angle QPA = 180^\circ$ --- (7)	Linear pair of angles.
8.	$\angle PSQ + \angle PTQ = 180^\circ$ --- (8)	Sum of opposite angles of a cyclic quadrilateral is 180° .
9.	$\angle QPB + \angle QPA = \angle PSQ + \angle PTQ$	From (7) and (8).
10.	$\angle QPB + \angle QPA = \angle QPB + \angle PTQ$	$\angle QPB = \angle PSQ$ from (6)
11.	$\angle QPA = \angle PTQ$	Hence (ii) is proved. This completes the proof.

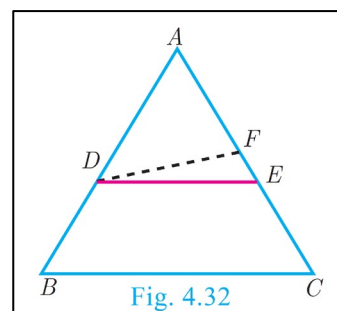
THEOREM 5**CONVERSE OF BPT (BASIC PROPORTIONALITY THEOREM)****1. STATEMENT**

❖ If a straight line divides any two sides of a triangle in the same ratio, then the line must be parallel to the third side.

2. GIVEN: In $\triangle ABC$, $\frac{AD}{DB} = \frac{AE}{EC}$

3. TO PROVE: $DE \parallel BC$

4. CONSTRUCTION: If DE is not parallel to BC , draw $DF \parallel BC$



S. NO	STATEMENT	REASON
1.	$\frac{AD}{DB} = \frac{AE}{EC} \dots (1)$	Given
2.	$\triangle ABC, DF \parallel BC$	Construction
3.	$\frac{AD}{DB} = \frac{AF}{FC} \dots (2)$	Thales theorem
4.	$\frac{AE}{EB} = \frac{AF}{FC}$ $\frac{AE}{EB} + 1 = \frac{AF}{FC} + 1$	From (1) and (2) Add 1 on both side
5.	$\frac{AE + EC}{EC} = \frac{AF + FC}{FC}$ $\frac{AC}{EC} = \frac{AC}{FC}$ $EC = FC$	Cancelling AC on both side
6.	Therefore, $E = F$ Thus $DE \parallel BC$	Our assumption that DE is not parallel to BC is wrong. Hence proved

THEOREM 6**CONVERSE OF ABT (ANGLE BISECTOR THEOREM)****1. STATEMENT**

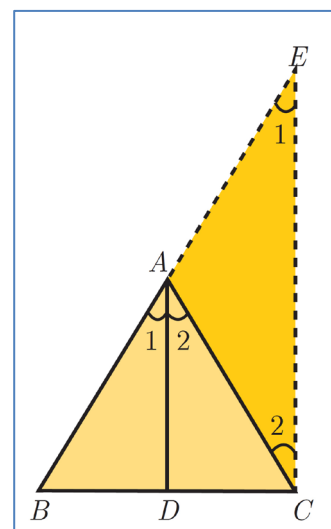
❖ If a straight line through one vertex of a triangle divides the opposite side internally in the ratio of the other two sides, then the line bisects the angle internally at the vertex.

2. GIVEN: ABC is a triangle. AD divides BC in the ratio of the Sides containing the angles $\angle A$ to meet BC at D.

$$\frac{AB}{AC} = \frac{BD}{DC} \dots (1)$$

3. TO PROVE: AD bisects $\angle A$ i. e. $\angle 1 = \angle 2$

4. CONSTRUCTION: Draw $CE \parallel DA$. Extend BA to meet at E.



S. NO	STATEMENT	REASON
1.	Let $\angle BAD = \angle 1$ and $\angle DAC = \angle 2$	Assumption
2.	$\angle BAD = \angle AEC = \angle 1$	Since $DA \parallel CE$ and AC is transversal, corresponding angles are equal
3.	$\angle DAC = \angle ACE = \angle 2$	Since $DA \parallel CE$ and AC is transversal, Alternate angles are equal
2.	$\frac{BA}{AE} = \frac{BD}{DC} \text{ --- (2)}$	In $\triangle BCE$ by Thales theorem
3.	$\frac{AB}{AC} = \frac{BD}{DC}$	From (1)
	$\frac{AB}{AC} = \frac{BA}{AE}$	From (1) and (2)
	$\frac{AB}{AC} = \frac{BA}{AE}$	
4.	$AC = AE \text{ --- (3)}$ $\angle 1 = \angle 2$	Cancelling AB $\triangle ACE$ is isosceles by (3)
5.	AD bisects $\angle A$	Since, $\angle 1 = \angle BAD = \angle 2 = \angle DAC$. Hence proved

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PREPARED & TYPED BY
Y.SEENIVASAN. M.SC, B.ED
MATHS TEACHER - 8489880553